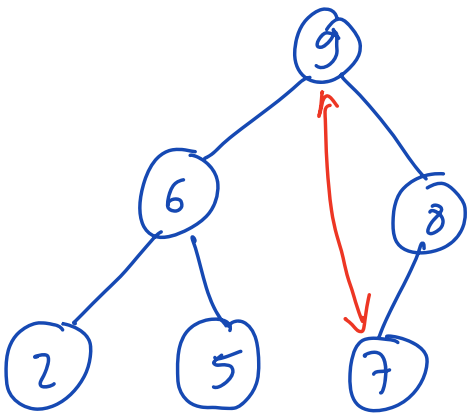
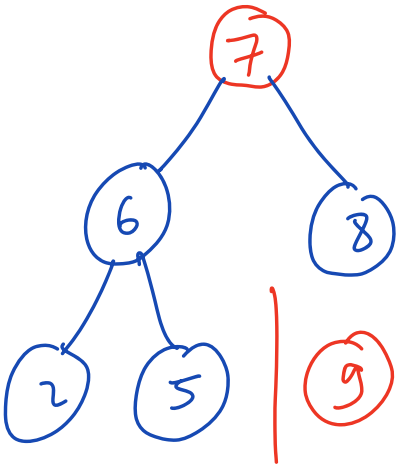
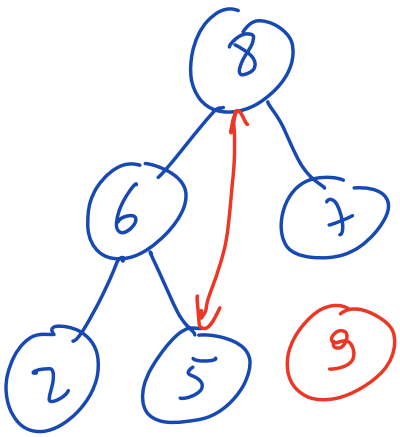
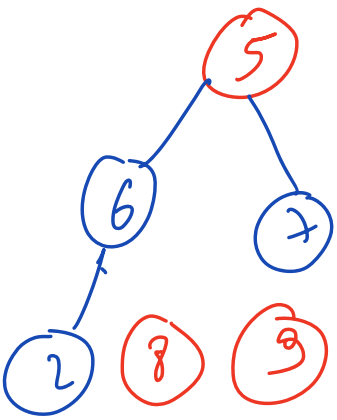
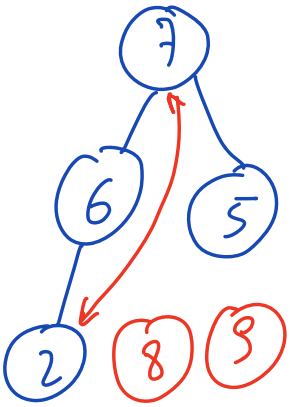
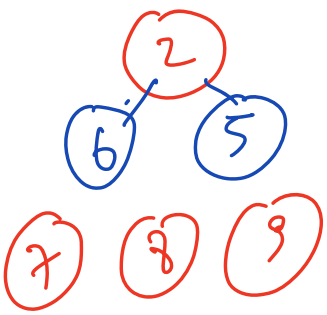

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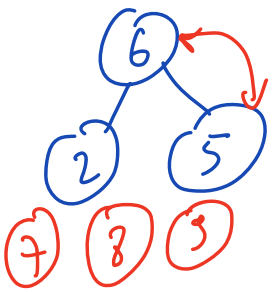

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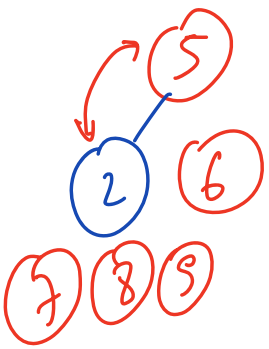

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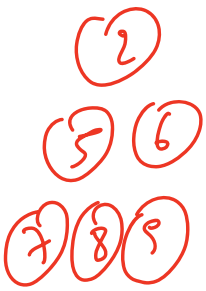

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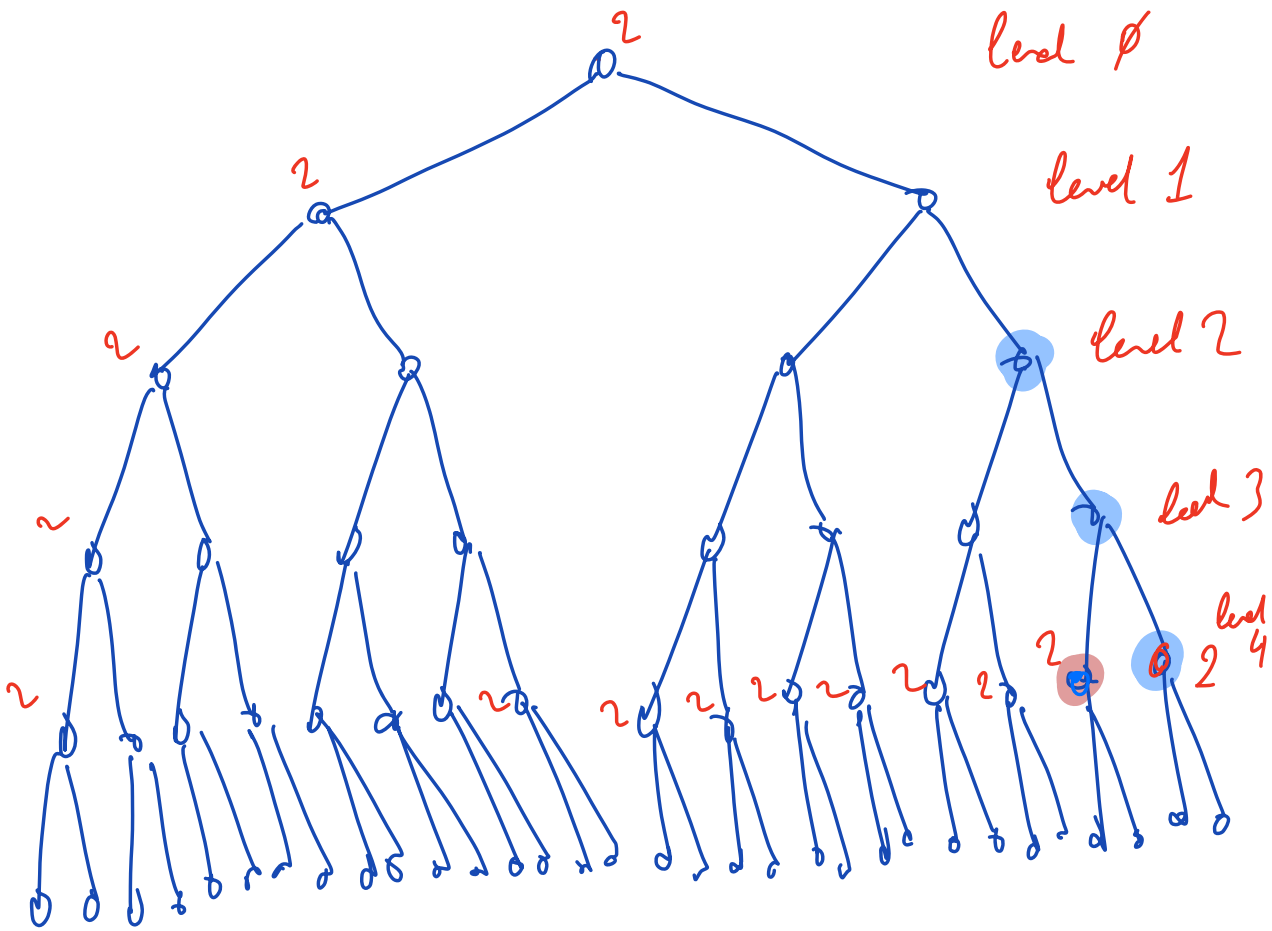

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$2^{h-1} * 2$ comparisons for level $h-1$

$2^4 * 2$ comparisons for level 4

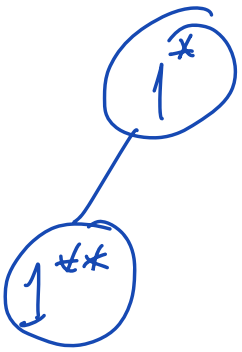
$2^3 \times 4$ " " level 3

$2^2 \times 6$ " " level 2

$2^1 \times 8$ " " level 1

$2^0 \times 10$ // // tied 0

1^* 1^{**}

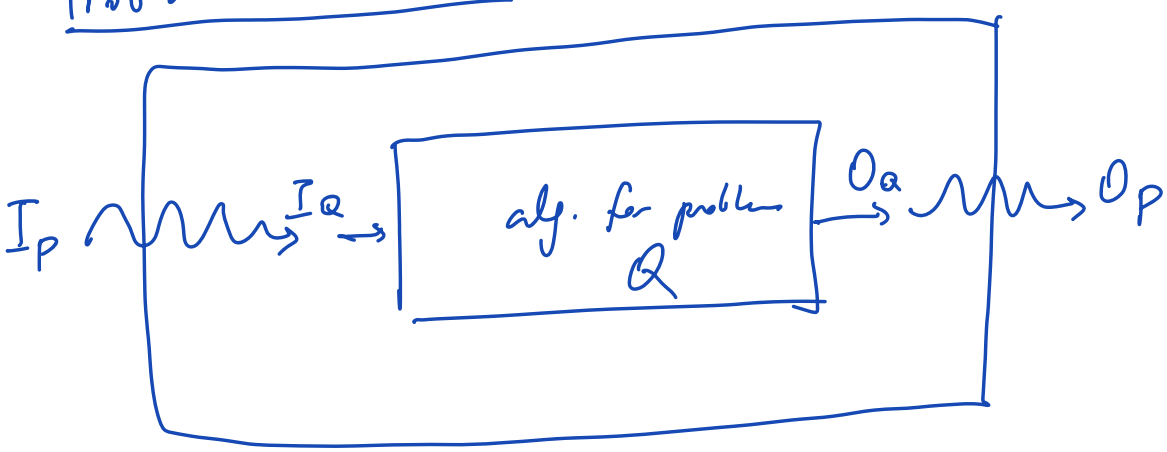


1^{**}

1^*

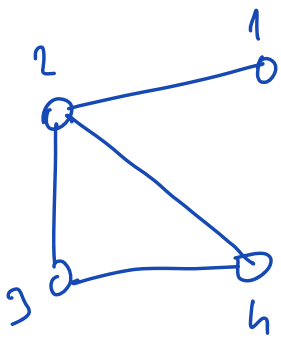
output 1^{**} 1^*

Problem Reduction

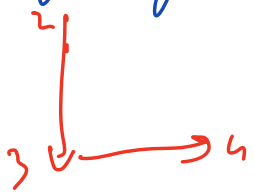


Q: gcd problem. alg.: Euclid's alg.

$$lcm(m,n) = \frac{m * n}{gcd(m,n)}$$

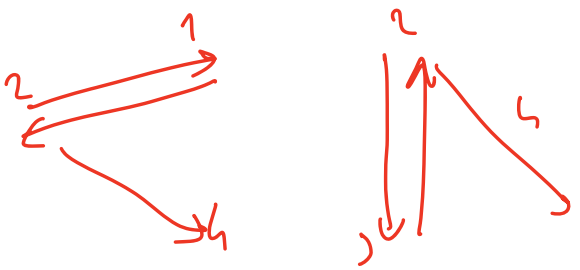


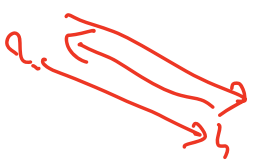
paths of length 2 from 2 to 4



1 path.

paths of length 3 from 2 to 4

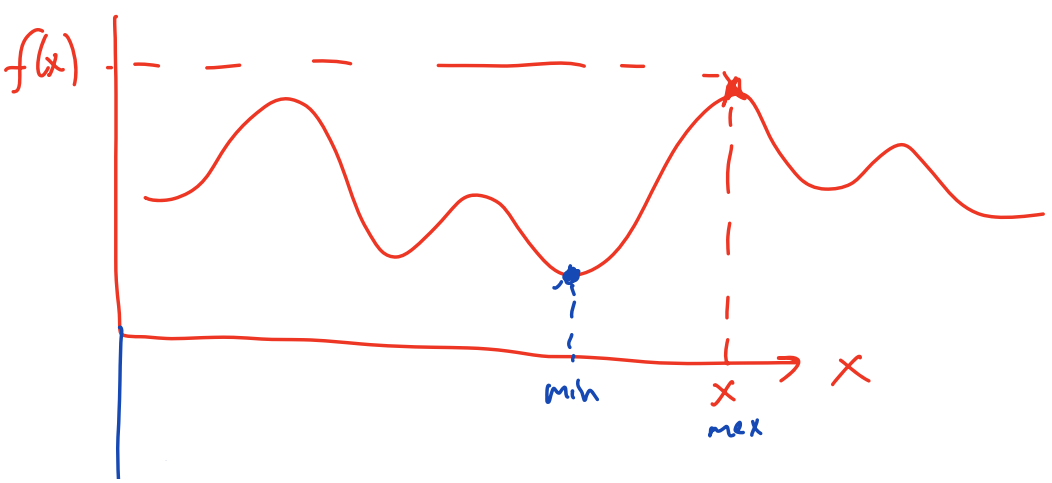




3 paths.

$$A = \begin{array}{c} \\ 1 \\ 2 \\ 3 \\ 4 \end{array} \begin{array}{c} \begin{array}{cccc} 1 & 2 & 3 & 4 \end{array} \\ \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{bmatrix} \end{array}$$

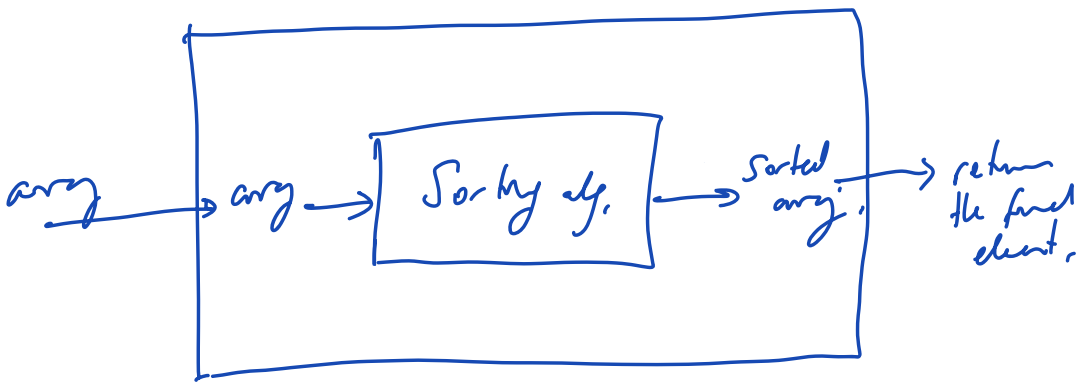
$A^2 =$





$$\max\{-f(x)\} = \min\{f(x)\}$$

Can we reduce Max Finding problem to Sorting problem?



But this reduction is not so efficient.

Linear programming: linear objective function.

$$\max \underbrace{3x + 5y - 2z}$$

$$\left.\begin{aligned} x &> y + z \\ z &> y - 2 \\ x + y + z &= 100 \end{aligned}\right\} \text{linear constraints}$$